

Supplementary Material for When Can We Trust Fairness Audits? Identifying Reliability Boundaries of Third-party Audit Conclusions

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A Proof of Proxy-band Concentration

This section supplements the proposition “Proxy-band Concentration” in the Methodology section. The main text uses this proposition to justify replacing the non-convex fairness constraint by a linear proxy $\mathbf{w}^\top \mathbf{h}$ when estimating the consistency radius.

We first restate the proposition using the notation of the main text. Let

$$\mathcal{H}^+ = \{\mathbf{h} \in \mathbb{R}_+^n : \mathbf{1}^\top \mathbf{h} = n, (1 - \mathbf{g})^\top \mathbf{h} > 0, \mathbf{g}^\top \mathbf{h} > 0\}$$

be the valid density-ratio domain on which the weighted group means in $\varphi(P_h)$ are well defined. For a fairness level ϕ , define

$$H_\phi := \{\mathbf{h} \in \mathcal{H}^+ : \varphi(P_h) = \phi\}.$$

Proposition 1 (Proxy-band Concentration). *For two points $\mathbf{h}, \mathbf{h}'$ sampled from H_ϕ and any vector $\mathbf{w} \in \mathbb{R}^n$, for any $\beta \in (0, 1)$,*

$$P\left(|\mathbf{w}^\top \mathbf{h} - \mathbf{w}^\top \mathbf{h}'| \leq \frac{A \cdot B}{\beta}\right) \geq 1 - \beta,$$

where

$$A = \sqrt{\mathbb{E}[\|\mathbf{h} - \mathbf{h}'\|^2]},$$

and

$$B = \sqrt{\mathbb{E}_{\mathbf{h}, \mathbf{h}' \in H_\phi} \left[\left(\int_0^1 \|\mathbf{w} - \nabla \varphi(P_{\mathbf{h}' + t(\mathbf{h} - \mathbf{h}')})\| dt \right)^2 \right]}.$$

Proof. Let

$$\gamma(t) = \mathbf{h}' + t(\mathbf{h} - \mathbf{h}'), \quad t \in [0, 1],$$

be the straight-line path from $\mathbf{h}'$ to $\mathbf{h}$. Since $\mathcal{H}^+$ is convex, this path remains in the valid density-ratio domain as long as both endpoint group masses are positive. Because the two endpoints lie in the same level set H_ϕ , we have

$$\varphi(P_h) - \varphi(P_{h'}) = 0.$$

Applying the fundamental theorem of calculus to $F(t) = \varphi(P_{\gamma(t)})$ yields

$$0 = \int_0^1 \nabla \varphi(P_{\gamma(t)})^\top (\mathbf{h} - \mathbf{h}') dt. \quad (1)$$

Now define the proxy discrepancy

$$Z(\mathbf{h}, \mathbf{h}') = |\mathbf{w}^\top \mathbf{h} - \mathbf{w}^\top \mathbf{h}'| = |\mathbf{w}^\top (\mathbf{h} - \mathbf{h}')|.$$

Using $\int_0^1 dt = 1$ and subtracting the zero-valued integral in Equation 1, we obtain

$$\begin{aligned} Z(\mathbf{h}, \mathbf{h}') &= \left| \int_0^1 (\mathbf{w} - \nabla \varphi(P_{\gamma(t)}))^\top (\mathbf{h} - \mathbf{h}') dt \right| \\ &\leq \int_0^1 \|\mathbf{w} - \nabla \varphi(P_{\gamma(t)})\| \cdot \|\mathbf{h} - \mathbf{h}'\| dt \\ &= \|\mathbf{h} - \mathbf{h}'\| \int_0^1 \|\mathbf{w} - \nabla \varphi(P_{\gamma(t)})\| dt. \end{aligned} \quad (2)$$

The inequality is the Cauchy–Schwarz inequality applied pointwise along the path.

Let

$$L(\mathbf{h}, \mathbf{h}') = \int_0^1 \|\mathbf{w} - \nabla \varphi(P_{\gamma(t)})\| dt.$$

Taking expectation over random pairs $(\mathbf{h}, \mathbf{h}') \in H_\phi \times H_\phi$ and applying Cauchy–Schwarz again gives

$$\begin{aligned} \mathbb{E}[Z(\mathbf{h}, \mathbf{h}')] &\leq \sqrt{\mathbb{E}[\|\mathbf{h} - \mathbf{h}'\|^2]} \sqrt{\mathbb{E}[L(\mathbf{h}, \mathbf{h}')^2]} \\ &= A \cdot B. \end{aligned} \quad (3)$$

Finally, $Z(\mathbf{h}, \mathbf{h}')$ is non-negative, so Markov’s inequality implies

$$P\left(Z(\mathbf{h}, \mathbf{h}') \geq \frac{A \cdot B}{\beta}\right) \leq \frac{\mathbb{E}[Z(\mathbf{h}, \mathbf{h}')] }{A \cdot B / \beta} \leq \beta.$$

Equivalently,

$$P\left(Z(\mathbf{h}, \mathbf{h}') \leq \frac{A \cdot B}{\beta}\right) \geq 1 - \beta,$$

which proves the proposition. $\square$

This proof also clarifies the role of the surrogate vector $\mathbf{w}$ in the algorithm. The bandwidth of the proxy values on one fairness level set is controlled by two factors: the effective diameter A of the level set and the gradient-misalignment term B . Since A is a property of the data geometry, the algorithm focuses on constructing a static $\mathbf{w}^*$ that reduces the misalignment with $\nabla \varphi(P_h)$ near the fairness boundary.

B Details of the Experiment in RQ1

This section supplements the RQ1 accuracy verification in the Experiments section. The main text reports the aggregated error on the Synthetic Trinomial dataset; here we provide the exact data construction, threshold selection, and ground-truth computation used for that comparison.

B.1 Synthetic Trinomial Construction

The synthetic environment is defined on a categorical random variable $X \in \{0, 1, 2\}$. The sensitive groups follow the main text: categories $X = 0$ and $X = 1$ form group D_0 , while category $X = 2$ forms group D_1 . Rather than explicitly defining a prediction model f , labels, and a performance function, we assign a performance value to each category:

$$\mathbf{m} = (m_0, m_1, m_2), \quad m_i \in [0, 1].$$

This represents the composite instance-level metric $M(f(x), y)$ used in the main paper.

For an audit distribution $\mathbf{p} = (p_0, p_1, p_2)$ and a shifted distribution $\mathbf{q} = (q_0, q_1, q_2)$ on the simplex, the fairness evaluation becomes

$$\varphi(\mathbf{q}; \mathbf{m}) = \frac{q_0 m_0 + q_1 m_1}{q_0 + q_1} - m_2, \quad (4)$$

where $q_0 + q_1 > 0$. We sampled 10 performance vectors $\mathbf{m} \in [0, 1]^3$ and 10 audit distributions $\mathbf{p}$ from a Dirichlet distribution with positive concentration parameters, producing 100 auditing configurations.

B.2 Ground-truth Radius

For each configuration $(\mathbf{m}, \mathbf{p})$ and threshold ϵ , the ground-truth consistency radius is the smallest KL shift that reverses the binary assessment:

$$R_{\text{gt}}(\mathbf{m}, \mathbf{p}, \epsilon) = \min_{\mathbf{q} \in \Delta_2} \sum_{i=0}^2 q_i \log \frac{q_i}{p_i}, \quad (5)$$

s.t. $\mathcal{A}(\mathbf{q}, \epsilon) \neq \mathcal{A}(\mathbf{p}, \epsilon)$.

where Δ_2 is the three-category probability simplex and $\mathcal{A}(\mathbf{q}, \epsilon)$ denotes the fair/unfair conclusion induced by $|\varphi(\mathbf{q}; \mathbf{m})|$ and ϵ .

Because the trinomial simplex is two-dimensional, we can enumerate possible shifted distributions by grid searching q_0 and q_1 and setting $q_2 = 1 - q_0 - q_1$. This exhaustive search gives a direct numerical ground truth against which CoRe and the baselines are compared.

B.3 Feasible Threshold Selection

The earlier version of this appendix described the feasible fairness range through a quadratic expression. Under the final camera-ready problem setting, that expression is unnecessary and can be misleading. With the group assignment used in the main text, the first term of Equation 4 is simply a convex combination of m_0 and m_1 . Therefore the attainable fairness values satisfy

$$\varphi(\mathbf{q}; \mathbf{m}) \in [\min(m_0, m_1) - m_2, \max(m_0, m_1) - m_2].$$

For each configuration, we only keep thresholds for which at least one shifted distribution can reverse the original assessment. From those valid thresholds, we choose 10 values and sort them by the audit margin

$$||\varphi(\mathbf{p}; \mathbf{m})| - \epsilon|.$$

The threshold ϵ_0 in the main table has the smallest audit margin and therefore corresponds to the most fragile assessment, while ϵ_9 has the largest margin.

B.4 Fine-grained Accuracy Pattern

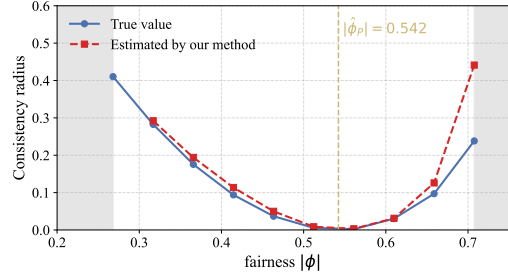


Figure 1: Ground-truth consistency radius and CoRe estimated radius for one representative Synthetic Trinomial configuration. The horizontal axis is the scanned fairness-boundary value, which corresponds to the tolerance threshold ϵ when evaluating assessment reversal for this configuration; it should not be confused with the margin-ordered indices $\epsilon_0, \dots, \epsilon_9$ in the main table.

Figure 1 shows one representative configuration before aggregation. CoRe tracks the ground-truth curve closely over most of the feasible threshold range. Larger deviations appear when the target boundary is near an edge of the probability simplex or far from the fair-anchor regime used to construct $\mathbf{w}^*$. This explains why the aggregate table in the main text shows the largest relative errors in the large-margin settings while retaining very small errors in the fragile, practically critical settings.

C Details of the Experiment in RQ2

This section supplements RQ2, which evaluates whether the estimated radius acts as a tight lower boundary for distribution shifts in settings where exact ground truth is unavailable.

C.1 Synthetic Gaussian

For the Synthetic Gaussian dataset, samples are drawn from $X \sim \mathcal{N}(0, 1)$. The sensitive groups are defined by the sign of X : samples with $X \leq 0$ belong to D_0 , and samples with $X > 0$ belong to D_1 . The performance value is set to

$$M(x) = \text{sigmoid}(x),$$

which creates a smooth performance disparity between the two groups without requiring an explicit prediction model and label.

To visually validate the radius as a lower boundary, we generate parametric distribution shifts by changing the mean and variance of the Gaussian distribution. Each shifted distribution induces a density-ratio vector $\mathbf{h}$ on the audit samples

and therefore a point $(|\varphi(P_h)|, d(\mathbf{h}))$ in the scatter plot. A valid radius curve should trace the lower envelope of these points, because the radius is the minimum shift required to reach each threshold.

C.2 Real-world Dataset and Model Settings

MovieLens-1M (Interaction Data). The MovieLens-1M dataset contains approximately 6,000 users and 4,000 movies. We remove users and items with fewer than five interactions and re-index the remaining IDs. We use a leave-one-out split: for each user, the most recent interaction is used for testing, and earlier interactions are used for training.

We train a LightGCN recommender with 64-dimensional user and item embeddings. Following the standard LightGCN design, graph convolution is performed without feature transformations or nonlinear activations. The model is optimized with Adam, learning rate 0.005, batch size 1024, and Bayesian Personalized Ranking (BPR) loss with one negative item sampled for each positive interaction. Early stopping is triggered if validation NDCG@10 does not improve for 5 consecutive epochs. The performance metric m_i is user-level recall.

Adult (Structured Data). For Adult, we train a multilayer perceptron to predict income level ($\leq 50K$ vs. $> 50K$). Missing values are removed. Categorical features are encoded with `LabelEncoder`, and numerical features are standardized with `StandardScaler`. The data are split into 80% training and 20% testing, with 20% of the training split reserved for validation.

The MLP has two hidden layers with 64 and 32 units, ReLU activations, and dropout rate 0.2. It is trained with Adam, learning rate 10^{-3} , and binary cross-entropy loss. Early stopping uses validation loss with patience 5. The performance metric m_i is prediction correctness.

CelebA (Image Data). For CelebA, the task is to predict the “Smiling” attribute. Images are center-cropped to 178×178 , resized to 128×128 , and normalized to $[0, 1]$. We train a ResNet-18 from scratch using Adam with learning rate 10^{-4} and batch size 64 for 20 epochs. The performance metric m_i is prediction correctness.

For all three real-world datasets, the sensitive attribute is sex (Male/Female), matching the main text.

C.3 Local Adversarial Validation

For each threshold ϵ , every radius-estimation method returns a critical shifted distribution $\tilde{\mathbf{h}}$ whose fairness evaluation is close to the boundary $|\varphi(P_h)| = \epsilon$. To test whether the estimated radius leaves blind spots near this boundary, we perturb $\tilde{\mathbf{h}}$ locally:

$$\mathbf{h}' = \Pi_{\mathcal{H}}(\tilde{\mathbf{h}} + \mathbf{z}),$$

where $\mathbf{z}$ is random noise with magnitude controlled by $\alpha \in \{0.01, 0.05, 0.10\}$ and $\Pi_{\mathcal{H}}$ projects the vector back to the valid density-ratio set $\mathcal{H} = \{\mathbf{h} \in \mathbb{R}_+^n : \mathbf{1}^\top \mathbf{h} = n\}$.

Let E_k denote the event that $\mathcal{A}(f, P_{h'_k}, \epsilon) \neq \mathcal{A}(f, P, \epsilon)$ and $d(\mathbf{h}'_k) \leq \hat{R}$. The Local Misjudge Rate (LMR) reported in the main text is

$$\text{LMR} = \frac{1}{K} \sum_{k=1}^K \mathbf{1}[E_k],$$

where $\hat{R}$ is the radius estimated by the corresponding method. A high LMR indicates that the claimed radius contains shifted distributions that already reverse the audit conclusion, which means the radius is overly loose.

D Details of the Experiment in RQ3

This section supplements RQ3, which studies how the estimated consistency radius changes with the audit sample size.

For each real-world dataset, we fix the trained model and vary only the audit set used for radius estimation. The audit sample size is varied from $n = 100$ to $n = 2,000$. For each sample size, we repeat the sampling and radius estimation 20 times and report the mean and standard deviation. The reference value in the main figure is computed with a larger audit sample ($n = 5,000$). This design isolates the sampling stability of CoRe from model training randomness.

The observed convergence around $n = 1,000$ should be interpreted as an empirical data-efficiency pattern rather than a universal sample-complexity guarantee. The required audit size can still depend on group imbalance, the spread of the performance values $\mathbf{m}$, and how close the audit assessment is to the threshold ϵ .

E Details for Addressing Practitioners’ Queries

This section supplements the Practical Implementation section, where the main text uses simulations to answer two questions raised by domain experts about how the consistency radius should be interpreted in practice.

E.1 Simulation Setup for Radius and Audit Margin

The first query asks whether the audit margin

$$||\varphi(P)| - \epsilon|$$

can serve as a proxy for the consistency radius. To examine this relation, we generate 3,000 synthetic audit datasets. Each dataset contains $N = 100$ samples with balanced binary groups: 50 samples in D_0 and 50 samples in D_1 . The instance-level performance values are sampled independently as

$$m_i \sim \mathcal{U}(0, 1).$$

For each dataset and threshold, we compute the consistency radius and the corresponding audit margin. We then quantify their association using Kendall’s rank correlation coefficient τ from `scipy.stats`. The main text reports a positive correlation, but this should not be read as a deterministic equivalence. The margin ignores the geometry of the shifted distribution space, while the radius explicitly minimizes the KL shift over that space.

E.2 Simulation Setup for Dataset Merging

The second query asks whether combining multiple audit datasets with the same fair/unfair conclusion necessarily yields a larger consistency radius. We construct 3,000 pairs of synthetic audit datasets $(\mathcal{D}_1, \mathcal{D}_2)$ using the balanced-group

procedure above. For a fixed threshold ϵ , we keep pairs whose two individual assessments agree.

For each retained pair, we compute three radii: r_1 for $\mathcal{D}_1$, r_2 for $\mathcal{D}_2$, and r_{merge} for the merged audit dataset

$$\mathcal{D}_{\text{merge}} = \mathcal{D}_1 \cup \mathcal{D}_2.$$

We compare r_{merge} with $r_{\min} = \min(r_1, r_2)$ and $r_{\max} = \max(r_1, r_2)$. The main text shows that merging often improves over the weaker individual radius ($r_{\text{merge}} > r_{\min}$), but it does not reliably exceed $r_{\max}$. Thus, agreement in binary audit conclusions is not sufficient to guarantee a stronger radius certificate.

F CoRe Algorithm Implementation Details

This section supplements the Consistency Radius Estimation Algorithm in the Methodology section by documenting implementation choices used in the experiments.

CoRe is implemented in Python. The inner optimization problem is solved with CVXPY. The KL objective is implemented through the convex entropy expression corresponding to

$$d(\mathbf{h}) = \frac{1}{n} \sum_{i=1}^n h_i \log h_i,$$

with constraints $\mathbf{h} \geq 0$ and $\mathbf{1}^\top \mathbf{h} = n$. The group-scale regularizer in the main text is used with $\lambda = 1.0$ by default. We use the ECOS solver for the exponential-cone program.

The outer binary search calibrates the proxy threshold ϵ_{proxy} until the true fairness value satisfies

$$||\varphi(P_h)| - \epsilon| < 10^{-5}$$

or the maximum number of iterations is reached. The maximum number of binary search iterations is 30, and the numerical tolerance used for solver-level termination is 10^{-4} .

When the original audit assessment is unfair, CoRe searches for the smallest shift that moves $|\varphi(P_h)|$ down to the fair boundary. When the original assessment is fair, assessment reversal can occur by crossing either $+\epsilon$ or $-\epsilon$; the implementation evaluates both boundary orientations when needed and returns the smaller feasible KL radius. This matches the definition of the consistency radius as the nearest shift that changes the binary assessment.

All experiments were conducted on a Linux server with an Intel Xeon Gold 6240 CPU and an NVIDIA GeForce RTX 3090 GPU. For audit datasets of size $n = 1,000$, CoRe typically converges within seconds, which is consistent with the binary-search structure described in the main text.